



OUR DAILY BREAD

Philippians 4: 13

"I can do everything through Him who gives me strength."

BIOLOGICAL SCIENCE (Cells)

LEVELS OF ORGANIZATION:

1. **Molecules**
 - smallest unit of most compounds
2. **Cells**
 - smallest functional unit of life
3. **Tissues**
 - group of cells
4. **Organs**
 - tissues grouped together to perform certain functions
5. **Organ System**
 - groups of organs which performs complex processes to sustain life
6. **Organism**
 - individual living thing
7. **Population**
 - group of organisms of one type which live in the same area
8. **Community**
 - populations which live together in a defined area
9. **Ecosystem**
 - community and its nonliving surroundings

CHARACTERISTICS OF LIVING THINGS:

- ☐ Made up of cells unicellular or multicellular
- ☐ Ability to reproduce
- ☐ Has genetic make-up except viruses
- ☐ Obtains energy and materials for living
- ☐ Responds to environment
- ☐ Maintains homeostasis (stable internal environment)

CELLULAR ORGANIZATION

- **Cell:** collection of living matter enclosed by a barrier that separates the cell from its surroundings.

Cell Theory

- ✓ All living things are composed of cells.
- ✓ Cells are the basic units of structure and function in the living things.

- ✓ New cells are produced from existing cells.

- **Robert Hooke** – coined the term "cells" after observing them from a slice of cork under the light microscope.
- **Matthias Schleiden** – concluded all PLANTS are made of cells.
- **Theodor Schwann** – concluded all ANIMALS are made of cells.

PROKARYOTES	EUKARYOTES
Smaller and simpler than cells of eukaryotes; have cell membranes and cytoplasm but no nuclei (ANUCLEATED); all bacteria are prokaryotes.	Aside from cell membrane and cytoplasm, they contain nucleus and other various organelles; all plants, animals and fungi are eukaryotes.

CELL STRUCTURES

1. Cell membrane

- Thin, flexible barrier that supports and protects the cells while allowing them to interact with their environment.
- It allows food to enter the cell and wastes to exit it.
- It is semi-permeable; certain substances readily pass through it such as water, alcohol and small lipids while some substances do not.
- **Lipid by-layer:** doubled-layer sheet which is the core of nearly all cells; also has protein channels embedded through the layer which serves as channels and pumps; some of these proteins have carbohydrate chains at the end which serves as identification markers.

2. Nucleus

- Contains the cell's genetic material and control the cell's activities.
- **DNA:** Deoxyribonucleic acid; the genetic material that contains the hereditary information of organisms.



- **Chromatin:** VISIBLE granular material within the nucleus with consists of DNA bound to protein
 - **Chromosomes:** CONDENSED form of Chromatin during cellular division
 - **Nucleolus:** small dense region inside the nucleus where assembly of ribosomes begins.
 - **Nuclear envelope:** double-membrane layer with NUCLEAR PORES to allow passage of materials in and out of the nucleus.
 - **Cytoplasm:** material inside the cell membrane excluding the nucleus.
- 3. Cell wall**
- To protect and provide support for the cell; found in plants, algae and most prokaryotes but NOT in animal cells.
 - **Cellulose:** most of plant cell wall is made up of this tough carbohydrate fiber
- 4. Cytoskeleton**
- A network of protein filaments that helps the cell to maintain its shape; also functions in moving the organelles within the cell.
- 5. Ribosomes**
- Small particles made of RNA (Ribonucleic acid) and protein which produces other proteins by following coded instructions.
- 6. Endoplasmic Reticulum**
- Organelle in which components of the cell membrane are assembled and some proteins are modified.
 - **Rough ER:** part of the endoplasmic reticulum where proteins are synthesized
 - **Smooth ER:** contains enzymes which perform specialized tasks such as lipid synthesis.
- 7. Golgi Apparatus**
- Attaches carbohydrates and lipids to proteins
- 8. Lysosomes**
- Filled with enzymes which break down lipids, carbohydrates and proteins from food to be used by the cells; breaks down organelles which are not functioning already; plant cells HAVE NO lysosomes.
- 9. Vacuole**
- Sac-like structure where water, salt, proteins and carbohydrates are stored; plant cells have a single large CENTRAL VACUOLE
- 10. Chloroplasts**
- Plant cell organelle which uses the energy from sunlight to make energy-rich food molecules through photosynthesis.
- 11. Mitochondria**
- Makes high-energy compounds that the cell can use.

CELLULAR FUNCTIONS

- 1. Diffusion** – passage of substances across the cell membrane which does not require energy, thus a PASSIVE process

- Substance flow from higher concentration to lower concentration (CONCENTRATION GRADIENT) until equilibrium is achieved (state where there is equal concentration of substances between two locations).
- 2. Osmosis** – the diffusion of water through a selectively permeable membrane; solvent flow from lower concentration of solute to higher concentration of solute until equilibrium is achieved (equal concentration of solutes).
- Fluid flows from hypotonic to hypertonic solutions across a semi-permeable membrane in an attempt to make both solutions isotonic.
 - **Osmotic Pressure** – greater pressure exerted on the hypertonic side of a selectively permeable membrane.
- HYPOTONIC SOLUTION:** solution with lower concentration of solute; placing an animal cell on a hypotonic solution causes the fluids inside the cell to move out of the cell into the solution, thus the cell shrinks or CRENATES.
- ISOTONIC SOLUTION:** solution with just right concentration of solute, placing an animal cell on an isotonic solution doesn't change its appearance.
- HYPERTONIC SOLUTION:** solution with higher concentration of solute; placing an animal cell on a hypertonic solution causes the fluids from the solution to move into the cell, thus it expands and then bursts, LYSIS.

FACILITATED DIFFUSION – movement of substances across the cell membrane that do not readily pass the lipid bi-layer; movement is facilitated by proteins in the lipid bi-layer serving as CHANNELS.

- **Active transport:** movement of substances across the cell membrane against the concentration gradient (from area of lower concentration to area of higher concentration); some movement requires energy through the proteins in the lipid bi-layer serving as PUMPS.
- **Endocytosis:** taking material into the cell through active transport by means of in folding or pockets of the cell membrane; when large amounts of materials are taking in PHAGOCYTOSIS occurs; when water is taking in PINOCYTOSIS occurs.
- **Exocytosis:** excretion of large amounts of materials from the cell.

CELL DIVISION

Mitosis – series of changes in the nucleus, involving an exact duplication of complete set of chromosomes and the separation of these chromosomes into TWO IDENTICAL sets of chromosomes; occurs in cells making up the body organs.



Replication – exact duplication of each single-stranded chromosome during the non-dividing period, resulting in double-stranded chromosomes.

Cyclins – proteins that regulate the timing of the cell cycle in eukaryotic cells.

Cancer cells – rapidly dividing cells that do not respond to the signals that would normally stop them from dividing.

Chromatids – individual strands of double-stranded chromosomes joined at the CENTROMERE.

MITOTIC DIVISION

1. **Interphase** – non-dividing period, the cell grows and replicates its DNA and centrioles.
2. **Prophase** – first and longest phase of mitosis, the chromatin condenses into chromosomes; the centrioles separate and form a spindle that helps separate the chromosomes. The nuclear membrane breaks down.
3. **Metaphase** – chromosomes line up across the center of the cell. Each chromosome is connected to a spindle fiber at its centromere.
4. **Anaphase** – sister chromatids separate into individual chromosomes and are moved apart.
5. **Telophase** – the chromosomes gather at opposite ends of the cell and lose their distinct shapes; two nuclear membranes form.
6. **Cytokinesis** – also called cytoplasmic division, separation of cytoplasm which occurs during or at the end of mitosis resulting in the formation of TWO daughter cells each containing an identical set of chromosomes.

PROCESS OF MEIOSIS: involves two separate and distinct divisions; process which involves reduction division resulting to one-half number of chromosomes; occurs in cells for sexual reproduction; in males meiosis results in four equal sized gametes called sperms while in females only one large egg cell results from meiosis and the other three cells called polar bodies are not involved in reproduction.

- **Diploid** – cells that have paired number of chromosomes
- **Autosomes** – type of chromosome from diploid cells that contains information on body characteristics
- **Sex chromosome** – type of chromosome from diploid cells that determines sex of organism; XX pair produces females while XY pair produces males.
- **Haploid** – cells that have $\frac{1}{2}$ number of chromosomes.

1. **Meiosis I** – results to two haploid daughter cells each with half the number of chromosomes as the original cell; covers Interphase I to Anaphase I
2. **Interphase I** – cells undergo DNA replication, forming duplicate chromosomes
3. **Prophase I** – each chromosome pairs with its corresponding homologous chromosome to form a tetrad
4. **Metaphase I** – spindle fibers attach to the chromosomes
5. **Anaphase I** – the fibers pull the homologous chromosomes toward opposite end of the cell
6. **Meiosis II** – occurs after meiosis I which covers Prophase II to Telophase II
7. **Prophase II** – the chromatin condenses into chromosomes
8. **Metaphase II** – the chromosomes line up in the middle of spindle fibers along the equatorial plate
9. **Anaphase II** – the sister chromatids separate and move toward opposite ends of the cell
10. **Telophase II** – results in four haploid daughter cells